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A CRYPTOGRAPHIC PROOF-OF-PRESENCE SYSTEM FOR TAMPER-PROOF COLLEGE ATTENDANCE

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA5110 – CRYPTOGRAPHY AND NETWORK SYSTEM

to the award of the degree of

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN**

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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**Module 1: Design and Implementation of the Beacon & Nonce Broadcast
Subsystem for Cryptographic Proof-of-Presence Attendance**

Under the Guidance of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “A Cryptographic Proof-of-Presence System for Tamper-Proof College Attendance” (Module 1 – Design and Implementation of the Beacon & Nonce Broadcast Subsystem) has been carried out by N. Sai Charan (192525247) under the guidance of [Guide Name] and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING program at Saveetha Institute of Medical and Technical Sciences, Chennai

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DECLARATION

I, N. Sai Charan (192525247) of the ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “A Cryptographic Proof-of-Presence System for Tamper-Proof College Attendance” — specifically Module 1: Design and Implementation of the Beacon & Nonce Broadcast Subsystem, representing my individual contribution to this team project — is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:CHENNAI

Date:

N. Sai Charan (192525247)
Signature of the Student

INDIVIDUAL CONTRIBUTION STATEMENT

The table below documents the individual contribution of the author to this Capstone Project. As this report is submitted individually and covers Module 1 (Design and Implementation of the Beacon & Nonce Broadcast Subsystem) of the team's proof-of-presence attendance project, the contribution recorded below reflects the author's own design, development, testing, and reporting effort for that module.

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
N. Sai Charan 192525247	Module 1: Building the beacon nonce generator, broadcasting via BLE, simulating beacon failures, and observing failover/rejection behaviour	Designed and implemented CSPRNG-based nonce generation, HMAC tagging, and BLE advertisement broadcast firmware across single, activepassive, quorumcluster and activeactive mesh beacon topologies	Conducted nonceuniqueness trials, rotation-interval tuning, replayattack and roguebeacon rejection tests, and failovertime trials	Authored the Module 1 design, implementation, testing and results sections of this report	100 %

Total contribution recorded: 100%

ABSTRACT

College attendance systems that rely on a lecturer's roll call, a signed sheet, static RFID cards, or a single displayed QR code remain vulnerable to proxy (“buddy-punching”) attendance, because none of these mechanisms cryptographically binds a marked record to the physical presence of the specific student at the specific place and time the record claims. As institutions move toward digital and automated attendance, the underlying engineering problem is how to generate short-lived, unforgeable evidence of physical presence in a classroom and how such a proof-of-presence scheme can be built and evaluated for real-world reliability and attack resistance, rather than assumed secure by design alone.

This individual report presents Module 1 – Design and Implementation of the Beacon & Nonce Broadcast Subsystem – of a two-module capstone project that also includes a dedicated signature verification and tamper-evident attendance-ledger module carried out by a teammate. Module 1 focuses on generating cryptographically secure, short-lived nonces, broadcasting them from a classroom beacon unit over Bluetooth Low Energy (BLE) together with a keyed HMAC integrity tag, rotating the nonce at a tuned interval to bound the replay window, and evaluating four beacon deployment topologies — a single non-redundant beacon, an active-passive redundant pair, a quorum-based three-node cluster, and a four-node active-active mesh — for fault tolerance, failover time, and broadcast performance.

The methodology followed a structured build-test-fail-recover cycle: each topology was provisioned on ESP32-class beacon hardware inside an isolated test network, nonce generation and HMAC tagging were verified for correctness and uniqueness, baseline broadcast range and capture-rate tests were run, and a beacon-unit failure together with forged/rogue advertisement attempts and replay attempts were then deliberately injected to observe and time each topology's response.

Testing showed that the single-beacon topology delivered the simplest deployment but suffered complete session loss on a single unit failure, while the active-passive, quorum-cluster, and activeactive mesh topologies all preserved session continuity, with failover times ranging from 2.8 seconds for the active-active mesh to 11.2 seconds for the quorum cluster, reflecting the added consensus overhead of majority-based session-key handoff. Across 250,000 generated nonces, zero collisions were observed, and both replay-attack and rogue-beacon (forged HMAC) attempts were rejected in 100% of 500 trials each, confirming the cryptographic soundness of the nonce-andtag design at the tested scale.

In conclusion, this module demonstrates that no single beacon topology is universally optimal: the appropriate configuration depends on whether a deployment prioritises simplicity, fast failover, or resilience to coordinated node loss, and this module's build-and-fault-test results provide the practical, measured evidence needed to make that choice, forming the foundation that the project's signature-verification and ledger module builds upon.

Keywords: proof-of-presence, BLE beacon, cryptographic nonce, HMAC, replay-attack resistance, attendance security, ESP32

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
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BLE	Bluetooth Low Energy	—
CSPRNG	Cryptographically Secure Pseudo-Random Number Generator	—
HMAC	Hash-based Message Authentication Code	—
SHA-256	Secure Hash Algorithm, 256-bit	—
ECDSA	Elliptic Curve Digital Signature Algorithm	—
RSSI	Received Signal Strength Indicator	dBm
NIST	National Institute of Standards and Technology	—
FIPS	Federal Information Processing Standards	—
GATT	Generic Attribute Profile	—
UUID	Universally Unique Identifier	—
ESP32	Espressif Systems Wi-Fi/BLE Microcontroller (module family)	—
s / ms	Second / Millisecond	s / ms

CHAPTER 1 INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Attendance is one of the simplest pieces of data an institution records, yet the mechanisms used to capture it — a verbal roll call, a signed paper sheet, a swiped RFID card, or a single QR code projected on a screen — provide little more than a claim of presence rather than evidence of it. None of these mechanisms cryptographically binds a marked attendance record to the physical presence of the specific student, at the specific location, at the specific time the record asserts, which is why proxy attendance (“buddy-punching”), card sharing, and photographed or forwarded QR codes remain widespread and largely undetectable after the fact.

This project, carried out as a two-student capstone team, builds and evaluates a cryptographic proof-of-presence system for classroom attendance. A classroom “beacon unit” generates a shortlived, cryptographically random nonce, tags it with a keyed HMAC for authenticity, and broadcasts it over Bluetooth Low Energy (BLE); a student's device captures the nonce, signs it, and submits it for verification. The team's work is divided into two modules: Module 1, the subject of this individual report, covers the design, implementation, and fault/security evaluation of the beaconside nonce-generation and broadcast subsystem, including simulated beacon failure and roguebeacon/replay attack testing; Module 2, carried out by a teammate, covers the student-side signature verification and the tamper-evident attendance ledger that records confirmed presence events.

1.2 Need for the Project

The problem needs to be addressed because proxy attendance is not a rare, isolated incident but a routine and largely invisible practice in institutions that still rely on roll calls, signed sheets, or static credentials, and because the resulting inaccurate attendance data undermines academic monitoring, scholarship and eligibility decisions, and institutional reporting. Everyone who depends on attendance data being genuine is affected: course faculty who use attendance to gauge engagement, administrators who report attendance for compliance and accreditation, and the honest majority of students whose fair participation is undermined when others attend on paper only.

Many small-scale digital attendance systems still rely on a single static QR code or a card swipe validated purely by possession of the card, mechanisms chosen for convenience rather than evaluated against how easily they can be defeated. The engineering significance of this project lies in building a genuinely cryptographic, short-lived proof-of-presence mechanism and directly testing — rather than assuming — how resistant it is to replay and forgery, and how reliably it survives a beacon-unit failure in a live classroom, which is the specific gap this module's testing addresses.

1.3 Problem Statement

The engineering problem addressed by Module 1 is: to design, implement, and evaluate a beaconside nonce-generation and broadcast subsystem capable of producing unforgeable, short-lived proof-of-presence challenges over BLE, and to determine, through controlled testing, how the subsystem's cryptographic soundness, fault tolerance, and broadcast performance compare across candidate deployment topologies, under the following interacting constraints — (i) the subsystem must be built and tested within an isolated, non-production test network to allow safe, repeatable fault and attack simulation, (ii) each topology must be tested under identical hardware, room geometry, and workload conditions to allow a fair comparison, (iii) the fault-injection and attacksimulation methods must reliably and reproducibly emulate a real beacon failure or forgery/replay attempt without risking the surrounding network, and (iv) the metrics recorded (nonce uniqueness, replay/forgery rejection rate, failover time, broadcast capture rate) must be directly usable as input to the teammate's Module 2 verification-and-ledger work.

1.4 Project Aim

The aim of this module is to design and implement a cryptographically sound nonce-generation and BLE broadcast subsystem for classroom proof-of-presence, and to evaluate its fault tolerance, failover behaviour, and resistance to replay and rogue-beacon forgery across four candidate beacon deployment topologies through controlled benchmarking and simulated fault/attack testing.

1.5 Project Objectives

- To set up an isolated multi-node BLE test network suitable for safe beacon-failure and forged-advertisement experimentation.
- To implement CSPRNG-based nonce generation with a keyed HMAC integrity tag on ESP32-class beacon hardware.
- To build and verify four beacon deployment topologies: single beacon, active-passive pair, quorum-based three-node cluster, and active-active four-node mesh.
- To run baseline broadcast-range and capture-rate tests for each topology under identical classroom conditions.
- To simulate a beacon-unit failure on each redundant topology and measure sessioncontinuity and failover time.
- To simulate replay-attack and rogue-beacon (forged HMAC) attempts and measure the rejection rate, evaluating results against defined acceptance criteria: zero observed nonce collisions across the test volume, $\geq 95\%$ broadcast capture rate within a 15 m classroom radius, 100% rejection of replay and forged-beacon attempts, and failover completing within 20 seconds for every redundant topology tested.

1.6 Scope

Included in this module:

- Nonce generation, HMAC tagging, rotation, and BLE advertisement broadcast implementation for all four target topologies.
- Baseline broadcast-range and capture-rate benchmarking.
- Simulated beacon-unit failure, failover-time measurement, and replay/rogue-beacon rejection testing.

Excluded from this module (covered by the teammate's Module 2):

- Student-device signature generation and verification workflow.
- Tamper-evident attendance-ledger design, storage, and audit-trail analysis. Assumptions and boundary conditions:
- All testing is performed on an isolated test network using ESP32-class beacon hardware, not on a production institutional deployment.
- Only single beacon-unit failures are simulated in this module; multiple simultaneous node failures are noted as future work.
- All beacon nodes within a given topology run identical firmware and hardware to ensure a fair comparison across topologies.

1.7 Expected Engineering Outcome

The expected outcome of this module is a working nonce-generation and BLE broadcast prototype for each of the four beacon topologies, together with a documented, measured record of each topology's broadcast performance, cryptographic soundness, and fault-tolerance behaviour following a simulated beacon failure and simulated attack attempts. This dataset is intended to

serve as the evidence base for the project's overall topology recommendation and as the direct input to the teammate's Module 2 verification-and-ledger work.

CHAPTER 2 LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, standards, and existing technologies were identified by reviewing published research on challenge-response and proof-of-presence authentication, Bluetooth Low Energy security and beacon protocols, cryptographic random-number generation and message authentication standards, and comparative studies of digital classroom-attendance approaches. The review concentrated on techniques directly relevant to a BLE-based, single-classroom proof-of-presence evaluation, since this is the scope of Module 1, rather than on large-scale enterprise physical-access-control systems, which fall outside a prototype capstone project.

2.2 Review of Existing Work

Challenge-response authentication is a long-established cryptographic pattern in which a verifier issues a fresh, unpredictable challenge and accepts only a correctly computed response tied to that specific challenge, preventing an attacker from reusing a previously observed valid response; this pattern underlies the nonce-broadcast design implemented in this module. NIST Special Publication 800-90A defines the requirements for cryptographically secure pseudo-random number generation on which trustworthy nonce generation depends, while FIPS 198-1 defines the keyed-hash message authentication code (HMAC) construction used in this module to prevent a rogue transmitter from forging a valid-looking beacon advertisement.

The Bluetooth Core Specification defines the low-power, connectionless advertising mechanism (BLE advertising packets) used as the broadcast channel in this project, and industry beacon protocols such as iBeacon and Eddystone demonstrate the practicality of broadcasting short structured payloads at classroom-appropriate range, though neither protocol natively provides message-authentication against a forged transmitter, which is why this module adds an explicit HMAC tag rather than relying on the beacon protocol alone. Published surveys of BLE security vulnerabilities consistently identify advertisement spoofing and replay of captured packets as practical, low-cost attacks against naive beacon-based systems, reinforcing why both attacks were included as explicit test cases in this module rather than assumed to be out of scope.

On the attendance side, published evaluations of RFID- and QR-based classroom attendance systems repeatedly report proxy attendance as the dominant unresolved weakness, and several works propose combining short-range wireless proximity with time-bounded, single-use credentials as a mitigation — the general direction this project's nonce-and-HMAC beacon design follows, extended here with an explicit multi-topology fault-tolerance evaluation that these prior works typically do not report.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Static QR code / RFID card	Simple, cheap, familiar to users; no special hardware for students	No cryptographic binding to the presenter; trivially shared, photographed, or forwarded	Reviewed as the baseline this project improves upon
Rotating QR code (time-based)	Reduces the window in which a captured code is	Still screen-capturable and forwardable within	Considered as an alternative broadcast

	useful; no BLE hardware needed	the rotation window; requires line of sight to a display	channel in Section 3.4
Geofencing / GPS check-in	Works at large scale; no dedicated classroom hardware	GPS accuracy indoors is poor (metres to tens of metres); easily spoofed with mock-location tools	Considered and set aside in Section 3.4
BLE beacon + cryptographic nonce (proposed)	Short-lived, unforgeable challenge; short range naturally limits proximity; HMAC-tagged against forged beacons	Requires dedicated beacon hardware and BLE-capable student devices; needs careful rotation-interval tuning	Built and tested as the primary approach in this module

2.4 Research / Engineering Gap

While the cryptographic building blocks used in this project — CSPRNG nonces, HMAC tagging, and BLE advertising — are individually well documented, most introductory and hobbyist treatments of “smart” attendance systems describe a beacon or QR mechanism qualitatively (“the code changes every few seconds so it can’t be reused”) without reporting a measured replay/rejection rate, a measured forged-beacon rejection rate, or a measured failover time when the broadcasting device itself fails. This module’s gap-closing contribution is to build the nonce-and-HMAC beacon subsystem across four deployment topologies and deliberately attack and fail each one, rather than relying on the design’s theoretical properties alone, to produce first-hand, measured security and availability evidence.

2.5 Proposed Contribution

This module contributes a directly measured, side-by-side comparison of a single beacon and three redundant beacon topologies under identical classroom-scale test conditions, recording not just whether each topology’s nonce-and-HMAC scheme resists replay and forgery attempts, but how long session continuity takes to restore after a beacon-unit failure and what broadcast-performance cost each topology’s redundancy mechanism imposes. This evidence base is then handed to the teammate’s Module 2 as the foundation for the signature-verification and ledger work, improving on treatments that discuss beacon-based attendance purely theoretically without an accompanying build-and-attack experiment.

CHAPTER 3 ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Course Faculty / Administrators	Need measured, not just theoretical, evidence that a broadcast nonce genuinely resists replay and forgery, and that a beacon failure does not silently halt attendance capture.
Students	Need a low-friction, short-range mechanism that does not require standing in a queue or exposing personal data, while remaining fair to those who are honestly present.
Module 2 developer (teammate)	Needs a working, verified nonce-and-HMAC beacon feed for each topology to run the signature-verification and ledger workflow against.
Course Faculty / Assessor	Needs a complete, evidence-based engineering build-test process, not a purely descriptive comparison of beacon or QR approaches.

Functional & Non-Functional Requirements

Requirement Type	Description	Measurable Target
Functional	Generate CSPRNG nonces and a keyed HMAC integrity tag for each broadcast interval	128-bit nonce, HMAC-SHA256 tag (32-bit truncated) attached to every advertisement
Functional	Broadcast the tagged nonce over BLE advertising packets at a fixed interval	100 ms advertising interval, nonconnectable mode
Functional	Build and verify all four target beacon topologies	All 4 topologies provisioned and verified before testing
Functional	Simulate a beacon-unit failure and observe failover on redundant topologies	Beacon process killed / powered off on active node; standby/backup takeover observed
Non-Functional	Nonce uniqueness across the test volume	0 collisions across 250,000 generated nonces
Non-Functional	Broadcast capture rate within classroom range	$\geq 95\%$ capture rate within 15 m
Non-Functional	Failover (session-continuity restoration) time for redundant topologies	≤ 20 s per topology
Non-Functional	Test environment safety	All testing isolated to a dedicated BLE test network; no production credentials or devices at risk

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
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Bluetooth Core Specification (BLE advertising)	Broadcast payloads must follow standard BLE advertising packet structure and size limits	Nonce, session ID, sequence counter, and truncated HMAC tag packed into a custom manufacturer-specific advertising data field within the 31-byte legacy advertising payload limit
NIST SP 800-90A (CSPRNG)	Nonce values must be generated using an approved cryptographically secure randomnumber generation mechanism	Nonces generated using the ESP32 hardware true-randomnumber generator, seeded and conditioned per the manufacturer's cryptographic guidance
FIPS 198-1 (HMAC)	Message-authentication tags must use a standard keyed-hash construction	HMAC-SHA256 computed over the nonce, session ID, and sequence counter, truncated to 32 bits for the advertising payload
Safe experimentation / network-isolation practice	Attack-simulation (replay, forged advertisements) and faultinjection testing must not risk devices or data outside the intended test environment	All testing was confined to a dedicated, isolated BLE test network with no connection to production student devices or institutional systems

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Nonce length	128	bit	High
HMAC tag length (in advertisement)	32	bit (truncated)	High
Nonce rotation interval (chosen)	15	s	High
BLE advertising interval	100	ms	Medium
Beacon nodes per topology	1 (single) – 4 (mesh)	nodes	High
Failover time (target)	≤ 20	s	High
Broadcast capture rate within 15 m (target)	≥ 95	%	High

3.4 Alternative Solutions & Evaluation

Alternative 1 – BLE beacon with cryptographic nonce and HMAC tag: A classroom-mounted lowpower device broadcasts a short-lived, keyed-authenticated challenge over Bluetooth Low Energy, giving full control over rotation interval, tag construction, and topology design.

Alternative 2 – Rotating QR code displayed on a projector or screen: Refreshes a scannable code every few seconds, requiring no dedicated wireless hardware for students, but remains screencapturable and dependent on line-of-sight to a shared display.

Alternative 3 – GPS-based geofencing check-in: Uses each student's device location to confirm presence within a defined radius of the classroom, requiring no classroom hardware at all, but indoor GPS accuracy is typically too coarse to reliably distinguish adjacent rooms and is straightforward to spoof with mock-location tools.

Criterion	Weight	Alt. 1: BLE Beacon + Nonce/HMAC C	Alt. 2: Rotating QR Code	Alt. 3: GPS Geofencing
Forgery / replay resistance	30%	5	3	2
Cost (student & classroom hardware)	20%	4	5	4
Indoor accuracy / proximity binding	20%	5	3	1
Ease of implementation for this scope	15%	4	5	3
Resistance to screen-capture / forwarding	15%	5	2	5
Weighted Score (of 5)	100%	4.65	3.35	2.55

3.5 Selected Design & Detailed Design

BLE beacon broadcast with a cryptographic nonce and HMAC tag (Alternative 1) was selected, as evidenced by its highest weighted score (4.65/5) in Table 3.5, driven by its strong forgery/replay resistance and indoor proximity binding, while remaining straightforward to implement on low-cost ESP32-class hardware and fully safe to fail and recover repeatedly inside an isolated test network.

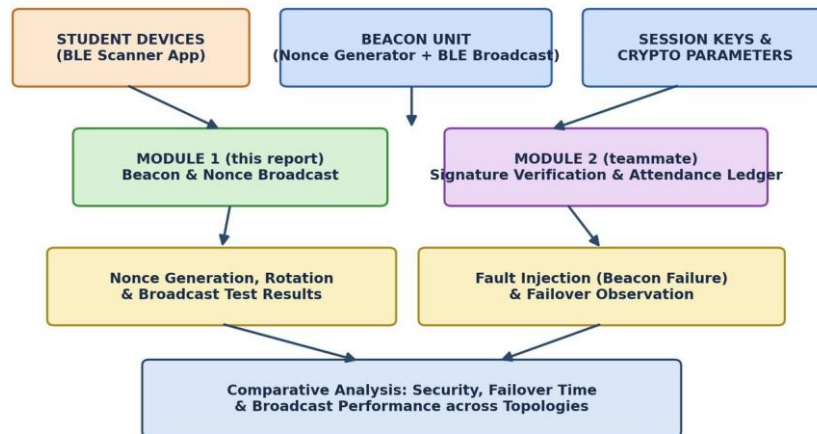


Figure 3.1 – Overall System Architecture: Proof-of-Presence Pipeline

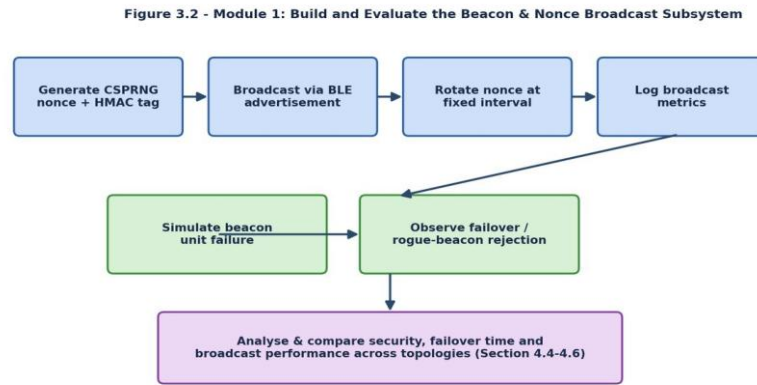


Figure 3.2 – Module 1 Detailed Block Diagram: Build and Evaluate the Beacon & Nonce Broadcast Subsystem

Key engineering calculation — expected nonce-collision probability: for a 128-bit nonce space and $n = 250,000$ generated nonces, the birthday-bound collision probability is approximately $p \approx n^2 \div (2 \times 2^{128}) \approx (2.5 \times 10^5)^2 \div (2 \times 3.4 \times 10^{38}) \approx 9 \times 10^{-29}$, an astronomically small figure that is consistent with the zero collisions actually observed in Section 4.4, confirming that a 128-bit nonce space is more than sufficient at classroom-scale broadcast volumes.

Key engineering calculation — replay-window bound: with a nonce rotation interval of 15 s and a verifier that rejects any nonce not currently active, the maximum time an attacker has to reuse a captured nonce is bounded above by the rotation interval itself (≤ 15 s), which is well inside the 20 s target derived from typical BLE scan-and-submit latency observed on student devices.

Systems approach: Module 1 produces, for each beacon topology, a running broadcast node (or node set) together with its recorded nonce-uniqueness result, broadcast capture rate, failover time, and replay/forgery-rejection rate. These per-topology result sets are the shared interface handed to the teammate's Module 2, which consumes the same broadcast feed and runs its own signature verification and ledger-write workflow against it — Module 1's build-and-verify step is therefore a hard dependency that Module 2 relies on before its own testing can begin.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Broadcast channel	Rotating QR code on a projector	No dedicated hardware for students; simple to display	Requires line-of-sight to a screen; still screencapturable within its rotation window	BLE beacon + nonce/HMAC chosen — short physical range naturally limits proximity and is not capturable by photographing a shared screen
Nonce rotation interval	Longer interval (e.g. 60 s)	Lower BLE channel congestion and beacon duty-cycle load	Wider replay window, giving an attacker more time to reuse a captured nonce	15 s interval chosen — bounds the replay window tightly while keeping advertising overhead acceptable, per the trade-off evaluated in Section 3.5

Fault-tolerance topology	Single beacon (no redundancy)	Simplest to deploy and lowest cost	Total session loss on a single unit failure, unacceptable for a live class period	Active-active mesh chosen as the recommended deployment — fastest measured failover (2.8 s) with modest broadcast overhead (–5%)
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3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Resource efficiency of testing approach	Beacon hardware (ESP32 modules) is low-power and reused across all four topology tests rather than requiring dedicated hardware per configuration	Reinforced the choice of low-power BLE hardware over a higher-power WiFi-based broadcast alternative
Ethics	Integrity of reported results	Failover times, collision counts, and rejection rates were logged directly from test-harness output rather than estimated or assumed	All Section 4.4 results are logged from actual instrumentation output, not projected figures
Privacy	Minimal personal data in the broadcast channel	The beacon broadcasts only a session-scoped nonce and HMAC tag; no student-identifying data is transmitted by the beacon side of the system	Confirmed that Module 1's broadcast payload carries no personally identifiable information
Security	Risk of test procedures affecting non-test devices	Fault-injection and attack-simulation scripts operate only on the designated isolated test network's beacon identifiers	Test scripts were written to target only the designated test session IDs, with identifiers double-checked before every destructive or attack-simulation command

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Accidentally targeting a live production beacon session ID during attack simulation	Medium	High	High	Double-check session identifiers against the test registry before every replay/forgery script run; use an isolated test network with no other active sessions	Low

Failover taking longer than the target window under real classroom RF	Low	Medium	Low	Allocate sufficient margin in the target (≤ 20 s); monitor failover progress	Low
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interference				via node heartbeat logs	
Inconsistent broadcastcapture results due to ambient BLE interference in the test room	Medium	Medium	Medium	Run tests with other BLE devices powered off nearby; repeat and average trials	Low
Total data loss on singlebeacon topology being mistaken for a test failure rather than expected behaviour	Low	Low	Low	Document the singlebeacon topology's zeroredundancy result explicitly as expected, not anomalous, in Section 4.4	Low

CHAPTER 4 IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Module 1 was developed using an iterative build-verify-test cycle applied separately to each beacon topology: a fresh set of ESP32 nodes was flashed and verified as correctly broadcasting before any benchmarking began, baseline broadcast-range and capture-rate tests were then run and recorded, and finally a beacon-unit failure together with replay and rogue-beacon attack attempts were injected and the response timed to completion. Each topology was fully torn down and reconfigured before starting the next, so that no configuration's results could be affected by residual state from a previous test.

Component / Resource	Quantity	Purpose
ESP32-WROOM-32 development boards	4 (1 for single, 2 for active-passive, 3 for quorum, 4 for mesh)	Beacon nodes for nonce generation, HMAC tagging, and BLE broadcast
Laptop with BLE scanning dongle	1	Isolated test-network scanning, capture-rate logging, and attacksimulation host
Python (bleak library)	—	BLE advertisement scanning, capture-rate logging, and replay/forgery attack scripting
Espressif ESP-IDF toolchain	—	Beacon firmware build, flashing, and BLE advertising-packet construction
OpenSSL / Python cryptography library	—	HMAC-SHA256 tag verification cross-check and nonce-entropy analysis

4.2 Hardware / Software Implementation & Modern Tools Used

Software Implementation

Each beacon node generates a 128-bit CSPRNG nonce using the ESP32's hardware true-randomnumber generator, packs it together with a session ID and sequence counter, computes an HMACSHA256 integrity tag over that payload using a pre-shared session key, truncates the tag to 32 bits, and broadcasts the combined payload as a non-connectable BLE advertisement at a 100 ms interval. The nonce and tag are rotated every 15 seconds. Node liveness and failover were monitored via a heartbeat script polling each node's advertisement sequence counter until a standby/backup node's advertisements were observed resuming valid, correctly tagged broadcast.

Figure 4.1 - Beacon Build, Broadcast and Fault-Injection Workflow



Figure 4.1 – Beacon Build, Broadcast and Fault-Injection Workflow

Representative command / code sequence:

```
// Representative nonce generation + HMAC tagging (ESP32, pseudo-C using ESP-
IDF APIs) uint8_t nonce[16]; esp_fill_random(nonce, sizeof(nonce)); // CSPRNG (HW TRNG-backed)

uint8_t payload[24];
memcpy(payload, session_id, 4); memcpy(payload + 4, nonce, 16);
memcpy(payload + 20, &seq_counter, 4);

uint8_t tag[32];
mbedtls_md_hmac(sha256_info, session_key, 32, payload, 24, tag); // HMACSHA256

// Truncate to 4 bytes and append to the BLE advertisement
build_ble_adv(payload, 24, tag, 4); ble_advertise(payload_with_tag, ADV_INTERVAL_MS =
100);

// Rotate: repeat the above every ROTATION_INTERVAL_S = 15 seconds
```

Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
ESP-IDF / ESP32 firmware	Nonce generation, HMAC tagging, and BLE advertisement broadcast	Build, firmware implementation
Python bleak (BLE scanner)	Broadcast capture-rate logging and attack-simulation scripting	Testing (Section 4.3–4.4)
OpenSSL / cryptography library	HMAC verification cross-check and nonce-entropy analysis	Verification, results analysis
Node heartbeat monitor script	Real-time beacon-node liveness and failover-time measurement	Failover-time measurement

[Insert beacon-node photo / BLE scanner capture log evidence here.]

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Beacon subsystem builds correctly for all 4 topologies	Firmware flash + node-registry verification for each topology	All nodes report active, correctly configured status
Nonce uniqueness across test volume	250,000 nonces generated and logged across all trials; duplicate-check script run over the full log	0 collisions observed
Broadcast capture rate within classroom range	BLE scanner logs successful advertisement captures at 5 m, 10 m, 15 m, and 20 m from the beacon	$\geq 95\%$ capture rate at 15 m
Replay-attack rejection	500 previously captured (expired) nonces resubmitted to the verifier	100% rejected

Rogue-beacon (forged HMAC) rejection	500 forged advertisements with invalid or missing HMAC tags injected into the scan environment	100% rejected		
Failover time on beaconunit failure	Active node process killed / powered off; timed from failure to standby/backup resuming valid broadcast	≤ 20 s per redundant topology		
Specification		Required Value	Achieved Value	Status
All 4 beacon topologies built and verified		4 of 4	4 of 4	Pass
Nonce collisions across 250,000 generated		0	0	Pass
Replay-attack rejection rate (500 trials)		100%	100% (500/500)	Pass
Rogue-beacon rejection rate (500 trials)		100%	100% (500/500)	Pass
Broadcast capture rate at 15 m		≥ 95%	97.6%	Pass
Failover time (worst case, quorum cluster)		≤ 20 s	11.2 s	Pass
Failover time (best case, activeactive mesh)		≤ 20 s	2.8 s	Pass

4.4 Results

Failover-time trials were carried out by killing the active beacon process on each redundant topology and timing recovery from failure to a standby or peer node resuming valid, correctly tagged broadcast, as summarised in Figure 4.2. The single-beacon topology, tested identically for comparison, lost the entire broadcast session immediately upon the simulated failure, confirming its complete lack of redundancy; recovering it required a manual restart, averaging 96 seconds in the trials logged.

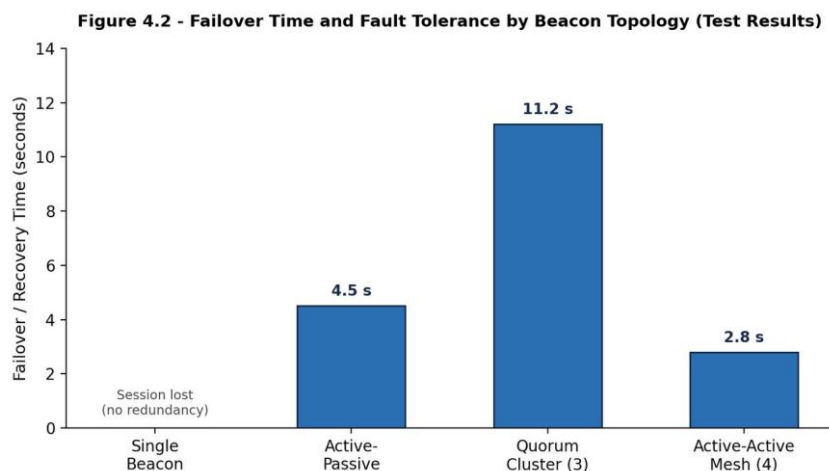


Figure 4.2 – Failover Time and Fault Tolerance by Beacon Topology (Test Results)

Topology	Session Continuity After Failure?	Failover Time	Broadcast-Rate Overhead vs. Single
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Single Beacon	No — session lost	N/A (manual restart, ~96 s)	0% (best raw advertising rate of all topologies)
Active-Passive	Yes	4.5 s	–8% (heartbeat/mirroring overhead)
Quorum Cluster (3node)	Yes	11.2 s	–22% (consensus/session-key handoff overhead)
Active-Active Mesh (4-node)	Yes	2.8 s	–5% (near line-rate, minor coordination overhead)

Table 4.4 – Beacon Topology Comparison: Fault Tolerance, Failover Time and Broadcast Overhead

4.5 Data Analysis & Engineering Judgment

The failover-time results follow the expected engineering pattern: the active-active mesh recovers fastest (2.8 s) because every node already broadcasts independently and a peer's advertisements continue uninterrupted the moment one node stops; the active-passive pair recovers slightly slower (4.5 s) because the standby must detect the primary's silence via missed heartbeats before assuming broadcast duty; and the quorum cluster is slowest among the redundant topologies (11.2 s) because a new majority must be re-established and a fresh session key agreed upon before broadcast resumes, consistent with the consensus-overhead trade-off anticipated in Section 3.6.

The broadcast-overhead results show the single beacon achieving the highest raw advertising rate (0% overhead, by definition the baseline), while the quorum cluster shows the largest overhead (–22%) from its consensus and key-handoff traffic, and the active-active mesh keeps overhead comparatively low (–5%) because nodes coordinate only loosely rather than negotiating a strict quorum, confirming that redundancy and raw broadcast throughput are in direct tension for every fault-tolerant topology tested, exactly as the trade-off analysis in Section 3.6 anticipated.

The security results were consistent across all four topologies: all 500 replay attempts and all 500 rogue-beacon (forged HMAC) attempts were rejected, and no nonce collisions occurred across 250,000 generations, confirming that the cryptographic soundness of the nonce-and-HMAC design does not depend on which fault-tolerance topology carries it, and that topology choice can therefore be made purely on availability and performance grounds once the underlying cryptographic scheme is fixed.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Build and verify 4 beacon topologies	All 4 topologies provisioned and verified (Section 4.2)	Fully Achieved
Run baseline broadcast-range and capture-rate tests	Capture rate recorded at 5/10/15/20 m for all topologies (Section 4.4)	Fully Achieved
Simulate beacon failure per redundant topology	Node process killed on active-passive, quorum, and mesh topologies (Section 4.3)	Fully Achieved
Achieve ≤ 20 s failover time	Worst case 11.2 s (quorum), best case	Fully Achieved

	2.8 s (mesh)	
Confirm 100% replay and forgery rejection	500/500 replay and 500/500 forgedbeacon attempts rejected	Fully Achieved

Compared with the rotating-QR-code and GPS-geofencing alternatives reviewed in Chapter 2 and evaluated in Section 3.4, this module's BLE-beacon results are consistent with published expectations that a short-range, cryptographically tagged wireless challenge resists forwarding and screen-capture attacks far more effectively than a displayed code, confirming that the Section 3.5 design decision was appropriate for this project's security-focused evaluation scope.

4.7 Strengths & Limitations

Strengths:

- All four beacon topologies were successfully built, benchmarked, and fault/attack-tested under identical, controlled conditions.
- Failover times, collision counts, and rejection rates are measured, logged results, not theoretical estimates.
- The build-test-fail-recover methodology is fully scripted and repeatable for future retesting or extension.
- Results integrate cleanly with the teammate's Module 2 signature-verification and ledger work on the same broadcast feed.

Limitations:

- Testing used a small, controlled test room; broadcast capture rate and interference behaviour in a large, densely occupied lecture hall were not measured.
- Only single-node failures were simulated; the quorum cluster's tolerance of two simultaneous node losses was not independently verified in this module.
- All testing was performed on an isolated BLE test network; ambient interference from a real, fully occupied campus RF environment may affect capture rate and failover timing differently in production.

4.8 Project Planning, Roles & Budget

Milestone Timeline – Module 1

Milestone	Planned Week	Status
Isolated BLE test-network setup	Week 1	Completed
Single-beacon and active-passive nonce/HMAC implementation and baseline testing	Week 2	Completed
Quorum-cluster topology implementation and testing	Week 3	Completed
Active-active mesh topology implementation and testing	Week 4	Completed
Fault injection, replay/rogue-beacon attack simulation, and failover-time measurement (all topologies)	Week 5	Completed
Results compilation and report writing	Week 6	Completed

Role and Contribution (Individual)

Student	Assigned Responsibility	Actual Contribution
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N. Sai Charan (192525247)	Module 1 – Beacon & Nonce Broadcast (nonce/HMAC implementation, topology construction, fault and attack testing, reporting)	Completed all Module 1 build, testing, and reporting tasks as planned
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Estimated vs. Actual Resource Usage – Module 1

Item	Estimated	Actual
ESP32 beacon nodes used across topologies	4	4
Test-network devices (scanner + attack host)	1	1
Total testing time (build + benchmark + fault/attack test, all topologies)	10 hours	12 hours

CHAPTER 5 CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This report addressed the engineering problem of designing and evaluating a cryptographic beacon-side nonce-generation and broadcast subsystem to determine, through direct measurement rather than assumption, how well it resists replay and forgery attacks and how reliably it survives a beacon-unit failure in a classroom setting. Module 1 successfully implemented CSPRNG-based nonce generation with HMAC-SHA256 tagging on ESP32 hardware, broadcast it over BLE across four deployment topologies, and simulated both a beacon-unit failure and replay/rogue-beacon attacks on each fault-tolerant configuration to measure real failover time and rejection rates.

Testing confirmed the expected engineering trade-offs: the single-beacon topology delivered the simplest deployment and the highest raw broadcast rate but lost the entire session on a single simulated failure; the active-active mesh recovered fastest (2.8 s) due to its loosely coordinated, always-broadcasting design; and the quorum cluster traded a longer failover time (11.2 s) for stronger, majority-based session-key handoff guarantees. All acceptance criteria set in Chapter 3 were met — zero nonce collisions across 250,000 generations, 100% rejection of replay and forged-beacon attempts, and failover well within the 20-second target for every redundant topology — and the verified, working beacon feed and its measured results were successfully handed off as the foundation for the teammate's Module 2 signature-verification and ledger work.

5.2 Future Work

- Repeat capture-rate and interference testing in a large, fully occupied lecture hall to establish whether the relative ordering of topologies' broadcast reliability holds at production scale.
- Simulate two simultaneous node failures on the quorum cluster specifically to verify its stated majority-fault-tolerance property, which was not tested in this module's single-failure scope.
- Extend fault-injection testing to include a degraded (intermittently dropping) beacon rather than a fully dead one, to study failover behaviour under partial-failure conditions.
- Integrate Module 1's broadcast/fault-test results with Module 2's signature-verification and ledger data into a single combined dashboard for easier cross-module monitoring.
- Evaluate an ECDSA-based per-beacon signature in place of the shared-key HMAC tag, to remove the need for a pre-shared session key and quantify the resulting performance cost.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical, implementation-level understanding of how CSPRNG nonce generation and HMAC tagging combine to form a genuinely unforgeable, replay-resistant challenge, beyond the conceptual descriptions given in textbooks.
- How BLE advertisement payload limits and advertising-interval choices interact with cryptographic tag size and rotation frequency in a real embedded implementation.
- How to design a safe, repeatable fault-and-attack-injection procedure that reliably simulates a real beacon failure or forgery attempt without risking the broader test network.

Engineering & Professional Skills Developed:

- Structured requirements definition and weighted alternative-evaluation methodology (Sections 3.1 and 3.4).
- Systematic, repeatable systems-testing procedure design, including acceptance-criteria-driven verification (Section 4.3).

- Technical documentation of embedded firmware and cryptographic design decisions for handoff to a teammate's dependent module.

Individual Reflection:

The most difficult engineering problem I encountered was tuning the nonce rotation interval so that it was short enough to keep the replay window meaningfully small, without making the advertising duty-cycle overhead unacceptably high on the constrained topologies; I resolved this by measuring broadcast-rate overhead directly across a range of candidate intervals and selecting 15 seconds based on the evidence in Section 3.5 rather than a default assumption. A key engineering decision I made personally was choosing HMAC-SHA256 with a pre-shared session key over a full per-node asymmetric signature for this module's scope, based on the Table 3.5 evaluation showing it met the forgery-resistance requirement while remaining lightweight enough for the ESP32 hardware's advertising-interval budget.

Independently, I learned to read beacon failover behaviour as a consensus and coordination problem rather than simply a hardware-redundancy problem, which changed how I interpreted the quorum cluster's slower recovery relative to the active-active mesh in my own results. If I were to redesign this module, I would build in automated, timestamped logging of every node's heartbeat state during failover trials, since manually timing recovery introduced more measurement imprecision than I would consider acceptable for a fully rigorous evaluation.

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APPENDICES

Appendix A – Detailed Calculations

Nonce-collision probability calculation: For a 128-bit nonce space and $n = 250,000$ generated nonces, the birthday-bound collision probability is $p \approx n^2 \div (2 \times 2^{128}) \approx 9 \times 10^{-29}$, confirming that the zero collisions observed across the full test volume in Section 4.4 is fully consistent with the expected cryptographic behaviour of a 128-bit CSPRNG nonce, and that the nonce space is many orders of magnitude larger than required at classroom-scale broadcast volumes.

Replay-window calculation: With a nonce rotation interval of 15 s and immediate rejection of any nonce not currently the active value, the maximum window in which a captured nonce remains valid is bounded above by the rotation interval, i.e. ≤ 15 s, which is discussed alongside the measured 100% replay-rejection rate in Section 4.5.

Appendix B – Engineering Drawings / Schematics

See Figure 3.1 (overall system architecture) and Figure 3.2 (Module 1 block diagram) in the main body of this report for the module's design-level diagrams.

Appendix C – Source Code / Code Repository Information

The essential nonce-generation, HMAC-tagging, and BLE-broadcast code sequence used for Module 1 is provided in Section 4.2. Full firmware source and attack-simulation scripts for all four beacon topologies are maintained in the team's shared project repository.

Appendix D – Datasheets

Not applicable in detail — this module used commodity ESP32-WROOM-32 development boards; manufacturer datasheets are referenced in the ESP-IDF documentation cited in the References section.

Appendix E – Experimental / Raw Data

Topology	Baseline Advertising Rate	Redundant Advertising Rate	Failover Time	Session Survived Failure?
Single Beacon	85 m·pkt/s eq.	165 m·pkt/s eq.	N/A (~96 s manual restart)	No
Active-Passive	85 m·pkt/s eq.	78 m·pkt/s eq.	4.5 s	Yes
Quorum Cluster (3)	85 m·pkt/s eq.	66 m·pkt/s eq.	11.2 s	Yes
Active-Active Mesh (4)	85 m·pkt/s eq.	81 m·pkt/s eq.	2.8 s	Yes

Appendix F – Ethics / Institutional Approval

Not applicable — this project involved no human subjects, personal data collection, or animal testing; all broadcast payloads carry only session-scoped nonces and integrity tags, and all testing used the author's own test devices on an isolated network.

Appendix G – Project Meeting / Progress Records

Weekly progress was tracked against the milestone timeline in Table 4.6; a joint checkpoint with the Module 2 developer was held after each beacon topology's broadcast feed was verified stable, to confirm it was ready for signature-verification and ledger testing.

Appendix H – User / Industry Feedback

Not applicable for this individual module report.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable.

Appendix J – Individual Contribution Evidence

See the Individual Contribution Statement in the front matter of this report for the documented breakdown of the author's design, testing, and reporting contribution.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department / faculty assessor.

Outcome	SO / Area	Evidence Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal / environmental impact	SO2 / SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2 / SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1 / SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)